How the confining branes are stabilized then gets intertwined with the Professor's struggle to understand and control gravitational anomalies. In the movie, that struggle is spelled out mathematically on the sixteen blackboards in Professor Brand's office; Chapter 25.



Fig. 23.8. A playing card, compressed end to end, bends and then buckles.

Traveling Through the AdS Layer



In the AdS layer, the AdS warpage of space produces tidal forces that are enormous by human standards. Any bulk being traveling through the layer to reach our brane must deal with those forces. Because we know nothing about the matter of which a bulk being is made—matter with four space dimensions—we have no idea whether this is an issue. In science fiction it can be left in the hands of the writers.

Not so for Cooper, riding in the tesseract (Chapter 29). In my interpretation of the movie, he has to cross the AdS layer. The tesseract must either protect him from the layer's enormous tidal forces or clear the AdS layer away from his path. Otherwise he'll be spaghettified.

By confining gravity, the AdS layer regulates its strength. In *Interstellar* we see gravity's strength fluctuate, perhaps due to